

K-MEANS CLUSTERING

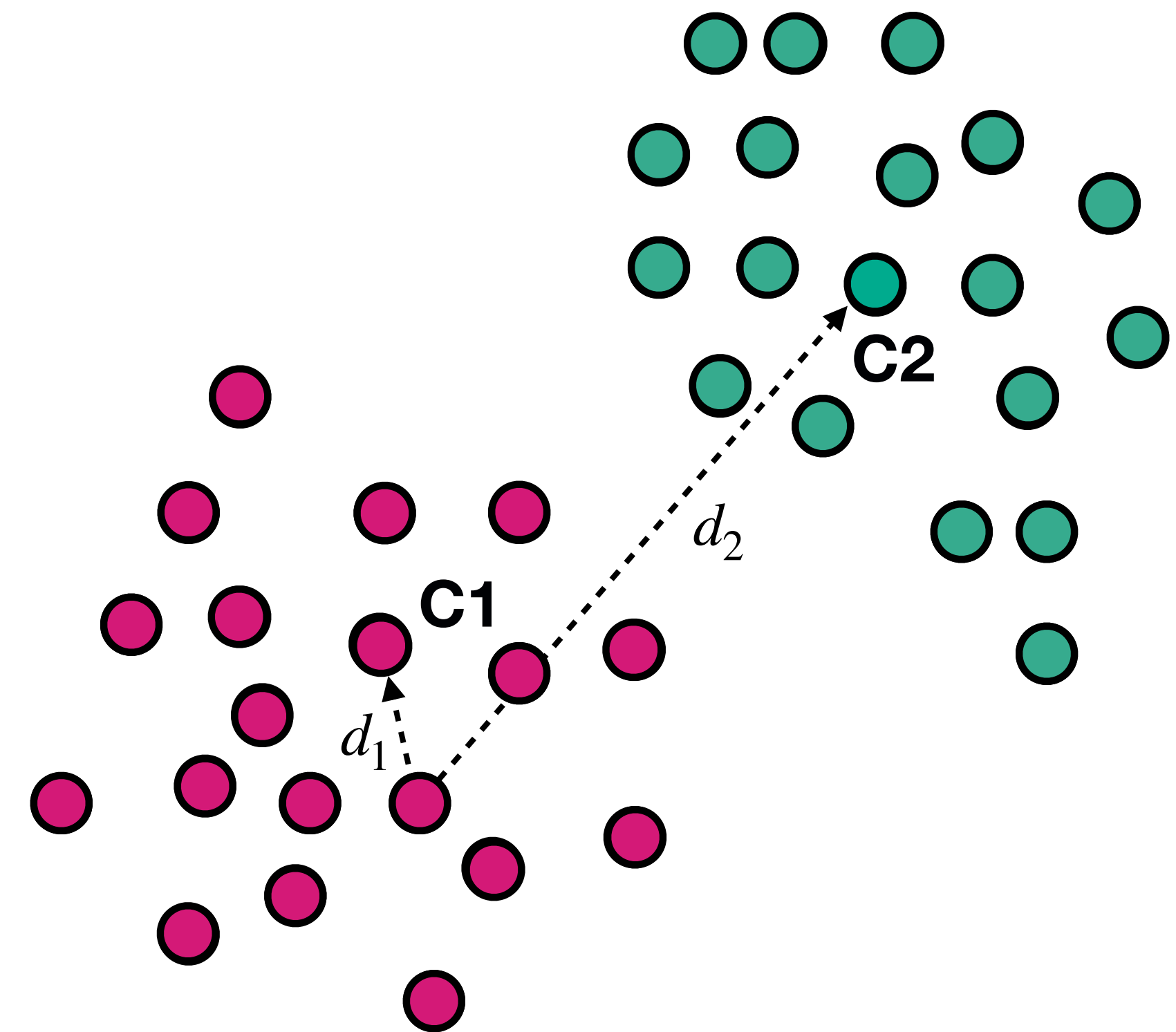
FROM NEIGHBOURHOODS TO GROUPING!

Objective– Partition n data points into K clusters such that the total within-cluster sum of square (WCSS) is minimized. Formally, the objective is to find:

$$\arg \min_S \sum_{i=1}^k \sum_{x \in S_i} ||x - \mu_i||^2^*$$

Where:

- S are the K clusters.
- μ_i is the centroid or mean of all the points in S_i .



* The $|| \cdot ||$ notation in formulas often denotes a norm, which is a function that assigns a positive length or size to each vector in a vector space, excluding the zero vector which is always assigned a length of zero.

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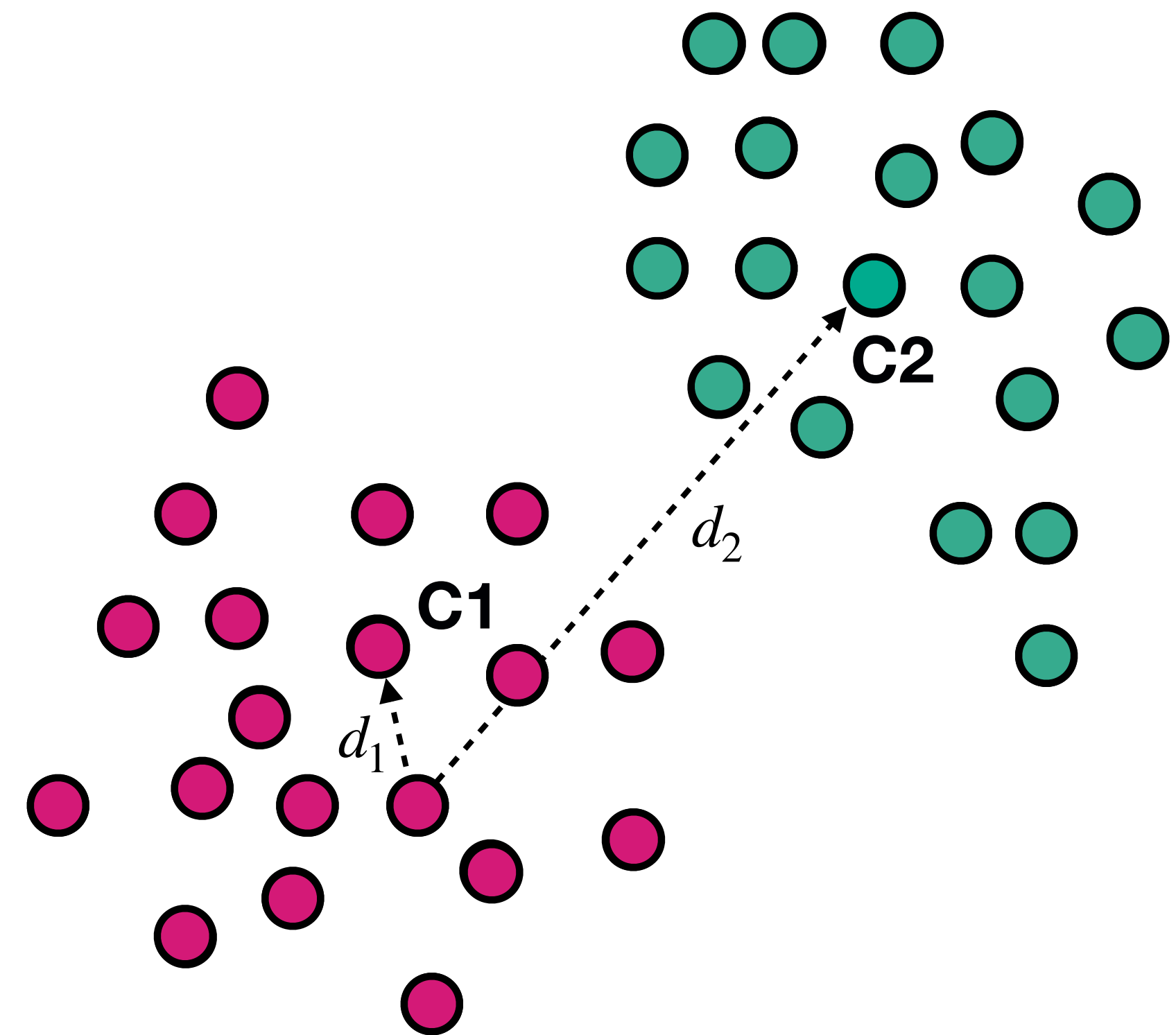
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